

Neural basis of social hierarchy across species

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Hierarch - What, why and how it varies

1. **Rank and relationships** — The highest-ranking chicken pecks all others; the second-ranking bird pecks everyone except the highest.
2. **Why hierarchies form** — Once mice establish dominance relationships, overt aggression declines.
3. **Different forms** — Female mouse hierarchies are generally less linear and less despotic than male hierarchies.
4. **Deep evolutionary roots** — Hierarchies occur in crickets, cichlids, chickens, mice and chimpanzees.
5. **Unequal benefits and costs** — Dominant male cichlids gain reproductive opportunities but have lower feeding and growth rates; subordinates are reproductively suppressed.

Hierarch - Three example routes to status

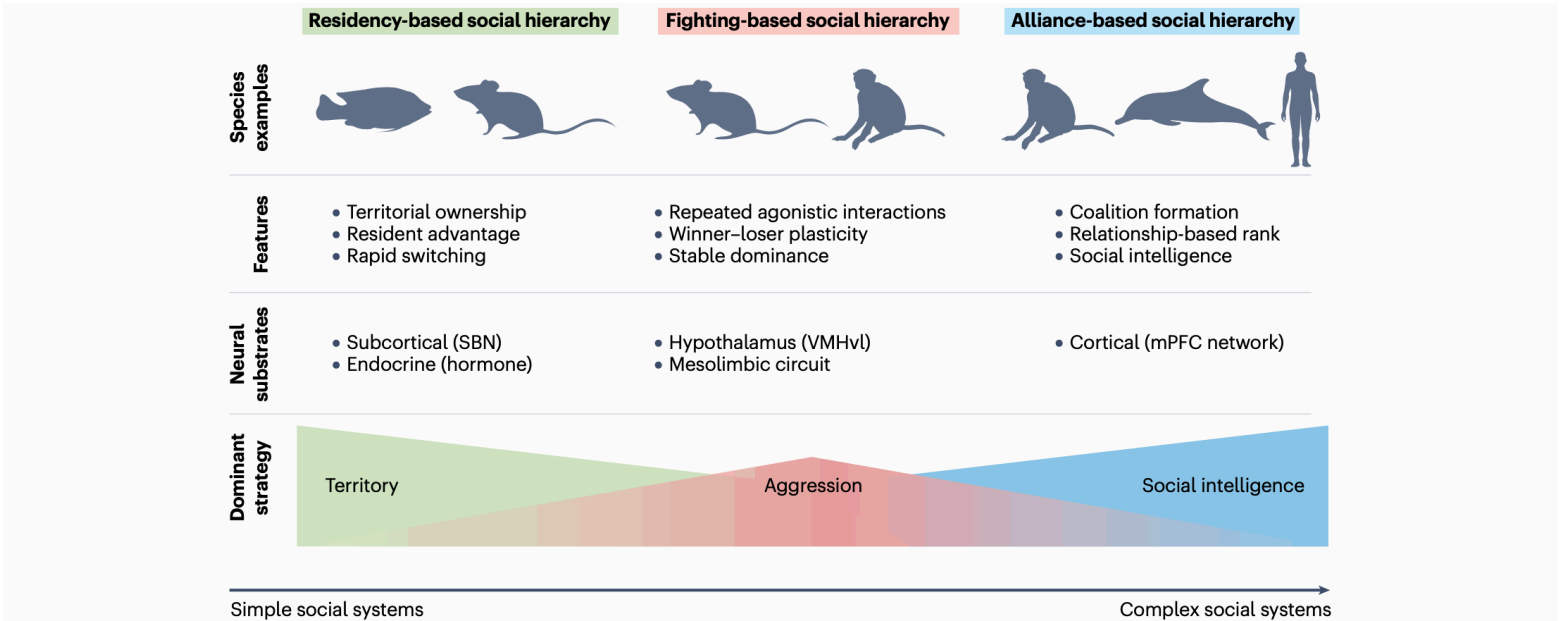


Fig. 4 | Evolutionary and neural organization of social hierarchy mechanisms. Social hierarchies arise through three major, partially overlapping mechanisms: residency-based, fighting-based and alliance-based strategies. In residency-based hierarchies, rank is determined by territorial ownership, with residents typically prevailing over intruders and status shifting rapidly as territory availability changes^{44,45,60–64}. This form of hierarchy is supported primarily by subcortical social behaviour network (SBN) and endocrine regulation^{46,66,70–72,74}. In fighting-based hierarchies, rank emerges through repeated agonistic interactions, during which winner-dependent and loser-dependent plasticity progressively biases animals towards aggression or avoidance, ultimately stabilizing dominance relationships^{1,33,79–81}. These processes are mediated by

hypothalamic circuits, including the ventrolateral part of the ventromedial hypothalamus (VMHvl)^{80,83,86,89,91}, together with a broader mesolimbic network^{103,104,109–112}. In alliance-based hierarchies, rank is shaped by coalition formation and social relationships, relying on strategic cooperation and social intelligence supported by cortical networks centred on the medial prefrontal cortex (mPFC)^{51,163–169,196,197}. Across evolution, as social systems increase in complexity and group size, there is a broad shift from territory-based and aggression-based strategies to alliance-based strategies, accompanied by a transition from predominantly subcortical to more cortical control^{176,177}. These mechanisms are not mutually exclusive, and many species combine multiple strategies to determine social rank.

Hierarch - The brain's role in hierarchy

1. Territory

1. **Social behaviour network (SBN)**: basic social behaviour circuits.
2. **Hypothalamic–pituitary–gonadal (HPG) axis**: reproductive hormone pathway.

2. Fighting

1. **Posterior ventrolateral ventromedial hypothalamus (pVMHvl)**: aggression.
2. **Anterior VMHvl (aVMHvl)**: defence and avoidance.
3. **Caudal medial preoptic area (cMPOA)**: suppression of aggression.
4. **Ventral tegmental area (VTA), nucleus accumbens (NAc), basolateral amygdala (BLA)**: reward and threat learning.

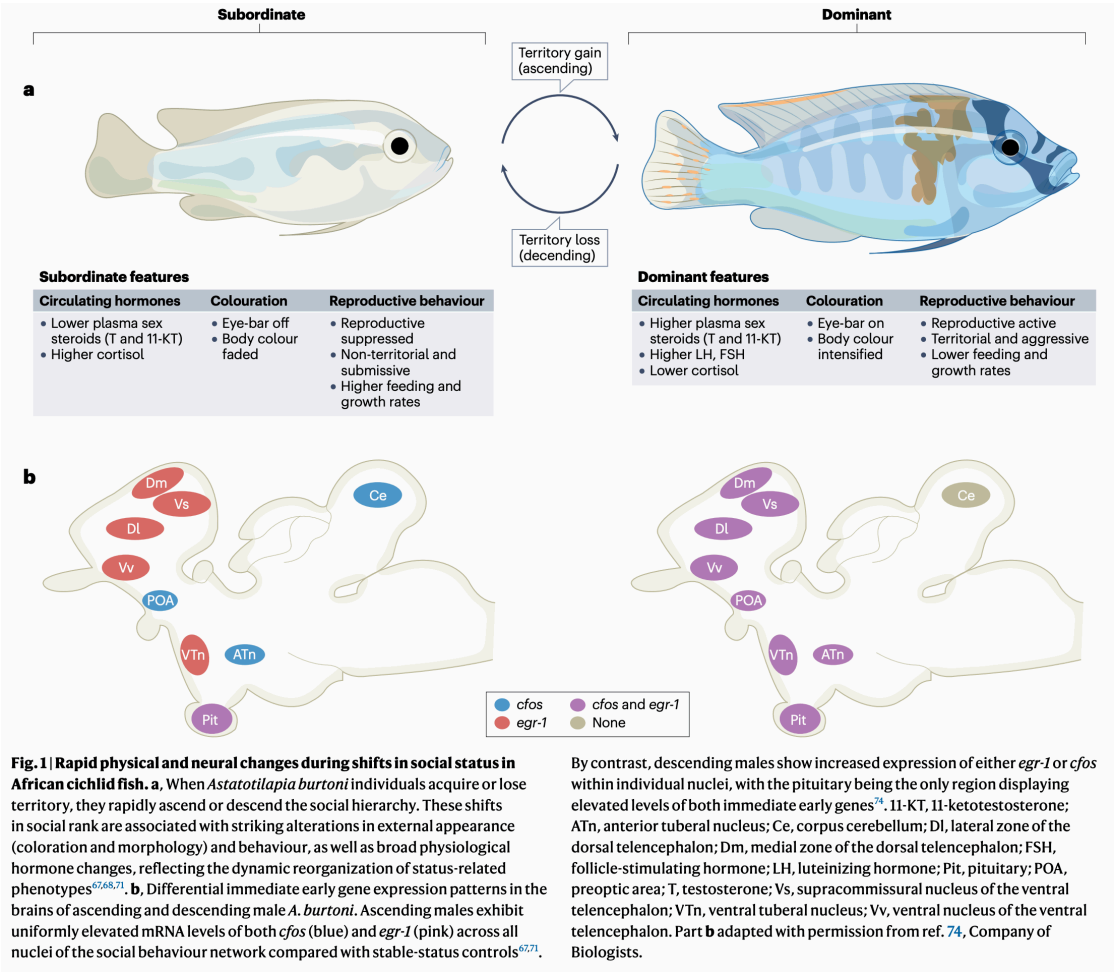
3. Social alliances

1. **Medial prefrontal cortex (mPFC), anterior cingulate cortex (ACC)**: social assessment and learning.
2. **Temporoparietal junction (TPJ), posterior superior temporal sulcus (pSTS)**: human mental-state inference and social cues.

Territory - Ownership gives residents an advantage

1. **Resident advantage** — In several bird species, residents tend to defeat intruders; body size alone does not determine the outcome.
2. **Territory enables dominance** — In African cichlid fish, territory holders defend spawning sites where females lay eggs, court females and reproduce.
3. **Ownership can change status** — Removing a dominant cichlid allows a subordinate to occupy its territory and begin defending it within minutes.
4. **Fighting still contributes** — Residents use chasing and threat displays; a vacant territory can trigger competition among several males.

Territory - A territorial opportunity changes behaviour



Territory - Hormones adapt to changing status

1. **Reproductive hormone pathway** — Status ascent activates the hypothalamic–pituitary–gonadal (HPG) axis, linking the brain to reproductive hormone production.
2. **Rapid ascent response** — Within 30 minutes, circulating sex steroids increase and steroid receptors are upregulated in hypothalamic regions involved in aggression and reproduction.
3. **Stable status differences** — Dominant male African cichlid fish maintain higher testosterone; subordinates have suppressed testosterone, elevated cortisol and reduced reproductive capacity.
4. **Response to displacement** — After a dominant male loses its territory, subordinate behaviour appears within approximately 30 minutes; cortisol rises within 24 hours.

Territory - The social behaviour network responds

1. **Distributed response** — Status transitions alter gene expression across the social behaviour network, including preoptic and hypothalamic regions.
2. **Ascending males** — The immediate early genes *cfos* and *egr-1* increase together across multiple network regions.
3. **Descending males** — Gene-expression changes follow a different, more selective pattern, including the preoptic area (POA) and anterior tuberal nucleus (ATn).
4. **Distinct transitions** — Gaining and losing status recruit different expression patterns; descent is not simply the reverse of ascent.

A reinforcing loop

Territorial opportunity → neural response → dominant behaviour → social feedback → further neural adaptation

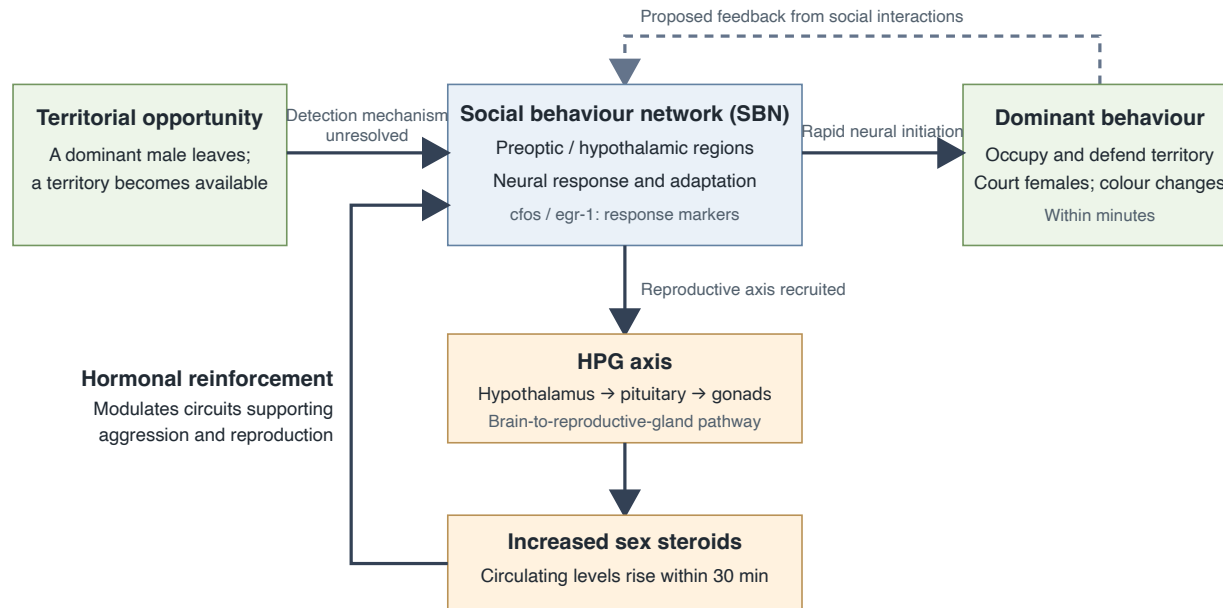
Hormonal changes ↔ neural circuits and behaviour

The review supports hormonal reinforcement and experience-dependent adaptation; the complete feedback loop is a synthesis, and the initiating mechanism remains unclear.

Territory - Neural and hormonal reinforcement

Territory: a reinforcing neural–endocrine loop

Male African cichlid fish (*Astatotilapia burtoni*)



Solid arrows summarize the review's account; they do not establish every causal step.

Dashed arrow: proposed feedback synthesis. Gene-expression markers do not identify the initiating circuit.

Based on Yan & Lin (2026), residency-based hierarchy section and Figure 1, pp. 497–498.

1. **Social behaviour network (SBN)** — Brain regions coordinating basic social behaviours, including aggression and reproduction.
2. **Preoptic and hypothalamic regions** — Brain areas linking social behaviour with bodily and hormonal responses.
3. ***cfos* and *egr-1*** — Genes whose increased expression marks a neural response.
4. **Hypothalamic–pituitary–gonadal (HPG) axis** — Hormone pathway linking the brain, pituitary gland and reproductive glands.
5. **Sex steroids** — Hormones, including testosterone, that influence reproductive function and behaviour-related circuits.
6. **Social feedback** — Consequences of interactions that may influence later behaviour and neural adaptation; the dashed loop is proposed.

Territory - How does opportunity become a status transition?

1. **Social opportunity** — Removing a dominant male makes its territory available; one or several subordinate males may respond.
2. **Rapid transition** — Within minutes, an ascending male can adopt dominant colouration, occupy the territory and defend it.
3. **Coordinated adaptation** — The review describes neural initiation followed by hormonal reinforcement, alongside changes in social behaviour network gene expression.
4. **Unresolved mechanism** — How the brain detects territorial opportunity and initiates these changes remains poorly understood.

Fighting - Fighting outcomes shape future encounters

1. **Winner effect** — Winning increases readiness to attack; after repeated victories, this can extend to other opponents.
2. **Loser effect** — Defeat reduces willingness to escalate subsequent aggressive encounters.
3. **Opponent-specific learning** — Defeated animals associate an opponent's cues with the aversive experience and avoid that individual.
4. **Hierarchy emerges over encounters** — Repeated interactions establish dominance relationships as animals adjust whom they challenge and whom they yield to.

Fighting - Winning strengthens aggression circuits

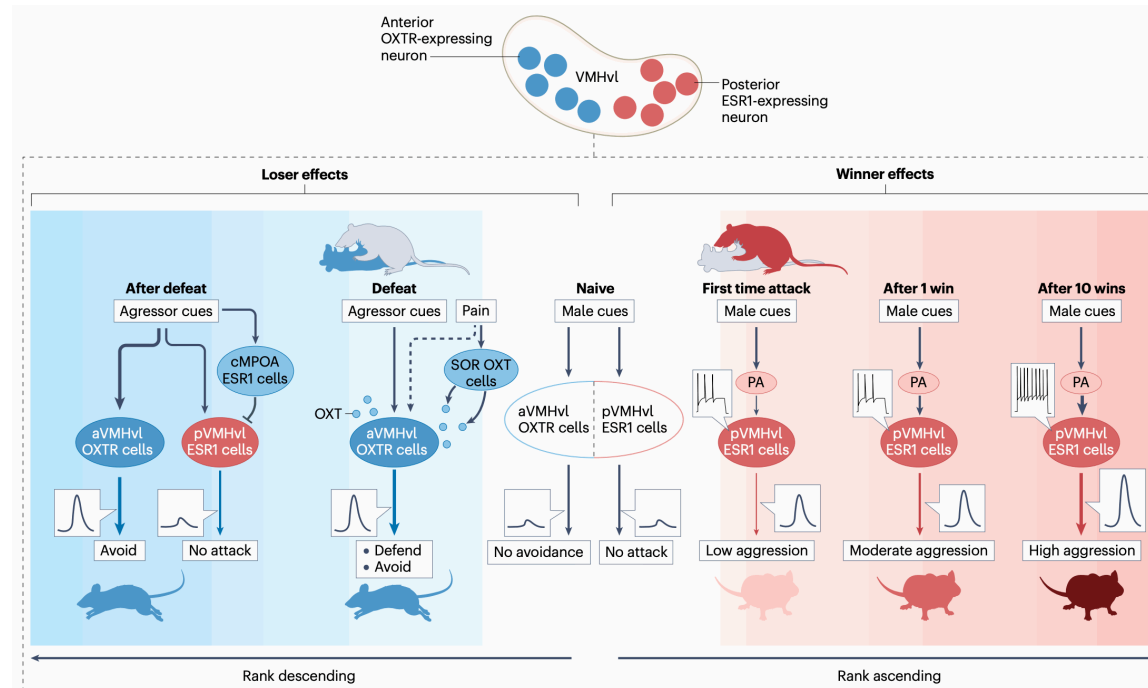


Fig. 2 | Hypothalamic circuits involved in fighting-based social hierarchy formation. Many mammals establish social rank through repeated agonistic interactions, with winner and loser effects encoded by distinct circuits centred in the ventrolateral part of the ventromedial hypothalamus (VMHvl). The neural mechanisms underlying behavioural adjustment after winning and losing in male mice are illustrated here. In naive, non-aggressive mice, the low activity of aVMHvl and pVMHvl neurons is insufficient to drive social avoidance or aggression. Repeated victories potentiate excitatory posterior amygdala (PA) inputs onto neurons in the posterior VMHvl (pVMHvl) that express the oestrogen receptor ESR1, increasing the excitability of these neurons and promoting winner-induced aggression and higher resource holding potential⁸⁰. By contrast, social defeat engages anterior VMHvl (aVMHvl) neurons expressing

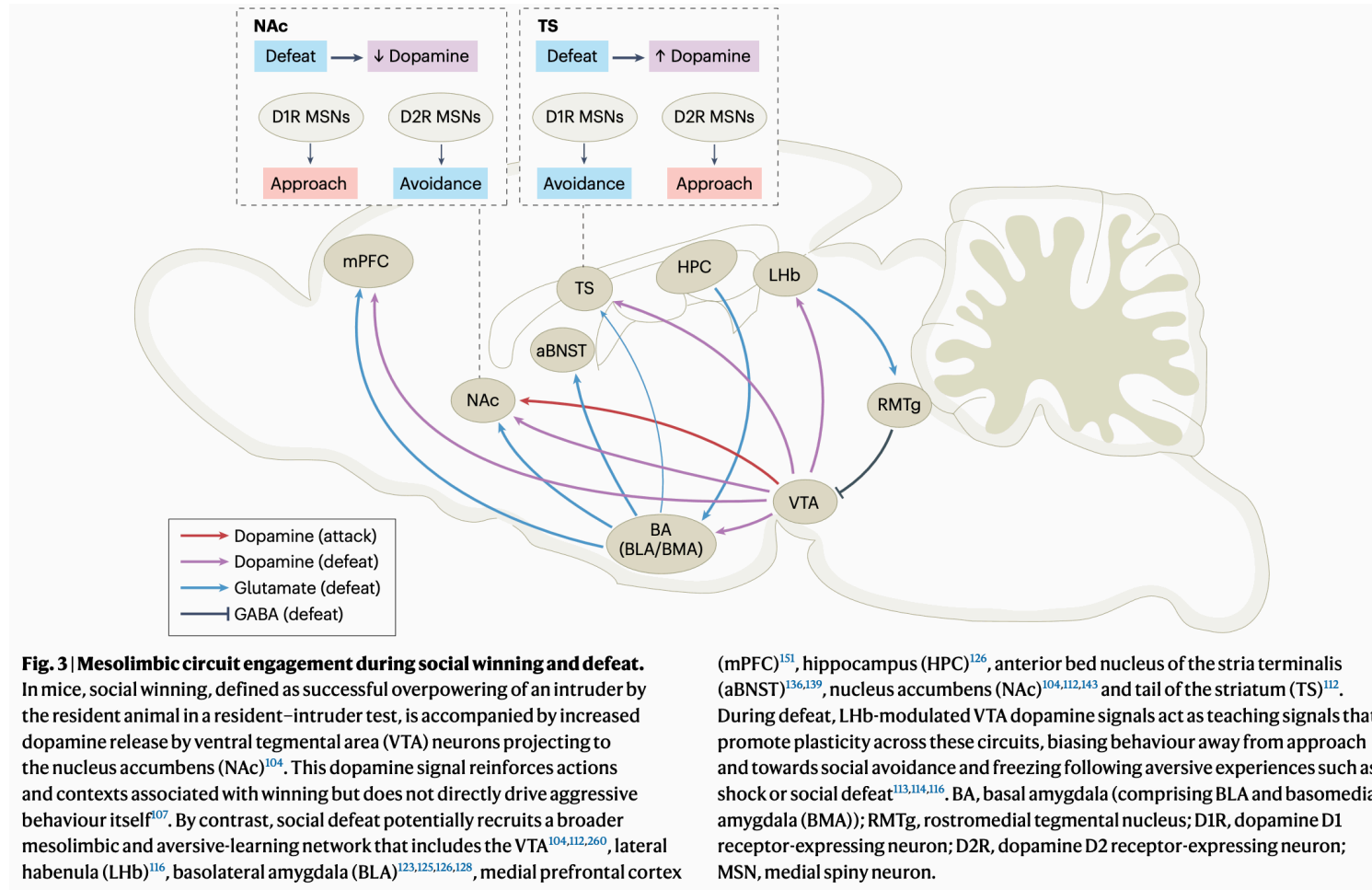
the oxytocin receptor OXTR: during defeat, pain activates neurons in the retrochiasmatic supraoptic nucleus (SOR), promoting oxytocin release, which strengthens aggressor-responsive synapses onto aVMHvl^{OXTR} cells, driving avoidance of dominant conspecifics⁸¹. aVMHvl activation during defeat also promotes active social defence⁹². After defeat, caudal medial preoptic area (cMPOA) neurons expressing ESR1 are also recruited, which suppresses the activity of ESR1-expressing pVMHvl cells and, thereby, reduces aggression towards higher-ranking opponents⁹¹. Together, these synaptic and cellular adaptations bias animals towards aggression or avoidance in response to specific social cues, helping to establish stable social hierarchies. Traces next to the blue and red arrows indicate outputs of the indicated pathways. Traces next to the red ovals indicate cell excitability.

Fighting - Losing suppresses attacks and teaches avoidance

1. **Suppressing aggression** — After defeat, aggressor cues recruit **caudal medial preoptic area (cMPOA)** neurons that inhibit aggression-driving **posterior VMHvl (pVMHvl)** neurons.
2. **Learning whom to avoid** — Pain during defeat triggers oxytocin release, strengthening aggressor-cue inputs onto oxytocin-receptor neurons in the **anterior VMHvl (aVMHvl)**.
3. **Opponent-specific memory** — A single 10-minute defeat can reduce investigation of, and proximity to, the aggressor the following day.
4. **Causal evidence** — Inhibiting the cMPOA neurons increases attacks on stronger opponents; activating the aVMHvl neurons induces avoidance even in undefeated mice.

VMHvl: ventrolateral part of the ventromedial hypothalamus. These findings concern mouse circuits.

Fighting - Defeat recruits broader learning circuits



Fighting - Defeat recruits broader learning circuits

1. **Teaching signals** — Dopamine from the **ventral tegmental area (VTA)** helps reshape responses to aggressor cues and defeat-associated contexts.
2. **Opponent and place memories** — The **basolateral amygdala (BLA)** supports learning about threatening social cues; the **ventral hippocampus (vHip)** contributes contextual information.
3. **Reduced approach** — Dopamine decreases in the **nucleus accumbens (NAc)** during attacks, favouring plasticity that reduces subsequent approach.
4. **Increased avoidance** — Dopamine in the **tail of the striatum (TS)** increases when defeated mice approach an aggressor, supporting threat avoidance.

Alliances - Social support can outweigh individual strength

1. **Coalition support** — Partners help individuals in conflicts and deter challengers, allowing physically modest individuals to attain high rank.
2. **Chimpanzees** — Physical attributes can influence initial rank, while enduring dominance is supported by alliances.
3. **Bottlenose dolphins** — Strong, stable alliances among largely unrelated males improve access to females and reproductive success.
4. **Relationships change rank** — Across several species, rank changes can follow shifts in coalition structure rather than changes in physical condition.

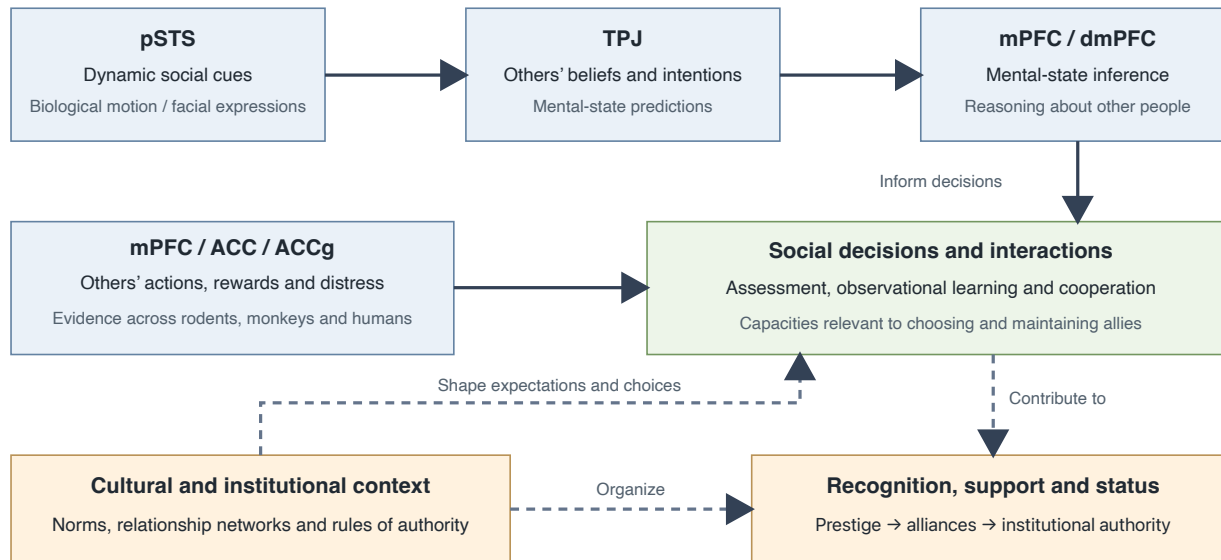
Alliances - Building alliances requires social intelligence

1. **Assess others** — Recognize individuals and track their behaviour, capabilities and emotional states.
2. **Learn by observation** — Rhesus monkeys can learn dominance relationships by watching interactions without participating.
3. **Adjust cooperation** — In cooperative tasks, dominant marmosets sometimes follow subordinate partners; leading and following need not match rank.
4. **Infer mental states** — Humans use theory of mind to anticipate others' beliefs, intentions and responses; comparable abilities in other species remain debated.

Alliances - Human neural networks and social systems

Human social cognition: interacting cortical networks

Functional synthesis of the review, rather than a complete anatomical wiring diagram



Solid arrows organize the review's functional account; they are not proofs of direct anatomical connections.

Dashed arrows: social-system interpretation. Neural capacities alone do not explain institutional authority.

Based on Yan & Lin (2026), pp. 497 and 504–506, and Table 1.

1. **Posterior superior temporal sulcus (pSTS)** — Processes movement and changing facial expressions.
2. **Temporoparietal junction (TPJ)** — Helps infer others' beliefs and intentions.
3. **Medial prefrontal cortex (mPFC)** — Supports social assessment and reasoning about others.
4. **Dorsomedial prefrontal cortex (dmPFC)** — Part of mPFC involved in mental-state inference.
5. **Anterior cingulate cortex / gyrus (ACC / ACCg)** — Represents others' distress and rewards; supports social learning. Evidence spans species.
6. **Norms and institutions** — Shared expectations and rules that organize support and authority; social systems, rather than brain regions.

Alliances - Neural substrates of social intelligence

Social intelligence functions	Region	Species	Refs.
Monitoring the actions and status of other individuals	PL, ACC	Mouse	189,201–204
	mPFC	Monkey	205,206
	mPFC	Human	207
Encoding the reward of other individuals (versus oneself)	ACC	Monkey	208,209
	dmPFC, VTA	Monkey	210
	NAc	Rat	229
Encoding the emotional state of other individuals	ACC	Mice or rat	211,212
	ACC	Vole	213
	ACC	Human	214,215
Effort evaluation and decision-making	ACC	Rat	216,219
	ACC	Monkey	217,218
Observation learning	ACC	Mouse	222
	ACC-BLA	Mouse	223
	mPFC	Monkey	206
Mental-state inference (ToM)	dmPFC	Human	234,235,238,239, 241,246,247
	vmPFC	Human	251
	pSTS	Human	237,238,241
	TPJ	Human	238,239, 248–250,252,253

1. **Prelimbic cortex (PL)** — Rodent frontal region involved in monitoring others.
2. **Anterior cingulate cortex (ACC)** — Supports social assessment, emotion processing and observational learning.
3. **Medial prefrontal cortex (mPFC)** — Frontal network representing information about oneself and others.
4. **Dorsomedial prefrontal cortex (dmPFC)** — Upper medial frontal region involved in social evaluation and mental-state inference.
5. **Ventromedial prefrontal cortex (vmPFC)** — Lower medial frontal region contributing to valuation and mental-state inference.
6. **Ventral tegmental area (VTA)** — Source of dopamine signals involved in learning.
7. **Nucleus accumbens (NAc)** — Region involved in reward and approach behaviour.
8. **Basolateral amygdala (BLA)** — Region linking cues to emotionally important outcomes.
9. **Posterior superior temporal sulcus (pSTS)** — Processes movement and facial cues.
10. **Temporoparietal junction (TPJ)** — Helps infer others' beliefs and intentions.
11. **Theory of mind (ToM)** — Inferring another person's mental state; an ability, rather than a brain region.

Alliances - Human status involves prestige, alliances and institutions

1. **Dominance and prestige** — Influence can arise through coercion or through respect voluntarily given for competence and achievement.
2. **Alliances amplify prestige** — The authors argue that expertise alone does not guarantee high status; relationship networks help sustain recognition and support.
3. **Institutions formalize authority** — Collective endorsement can turn prestige and social support into roles such as executive or director.
4. **Authority reinforces status** — Formal positions increase visibility and access to networks, helping sustain further influence.

Authors' proposed cycle: prestige → alliances → institutional status → expanded visibility and networks.

Interpretations - Interpretations

The review's human neural evidence mainly explains capacities for navigating social relationships, while its account of institutions leaves room for our interpretation that collective action can organize those capacities towards different distributions of status and authority.